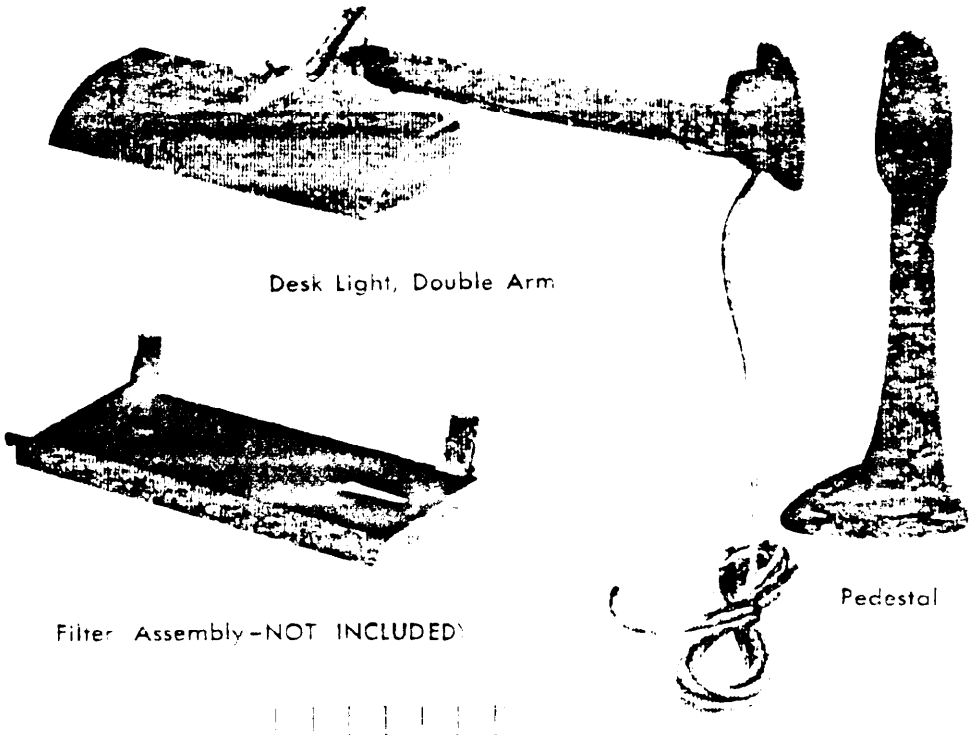
MILITARY SPECIFICATION SHEET

FIXTURES, LIGHTING; FLUORESCENT, DETAIL LIGHTING FOR DESKS

8 WATTS, 120 VOLTS, 60 HERTZ, SYMBOLS 141.2 AND 150.2

This specification sheet is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the fixtures described herein shall consist of this document and the latest issue of MIL-F-16377(SHIPS).



SH0004

FIGURE 1. Typical view of Symbols 141.2 and 150.2

All dimensions in inches.
For tolerances, other than specified herein, see paragraph 3.5.2 of MIL-F-16377.

FSC 6230

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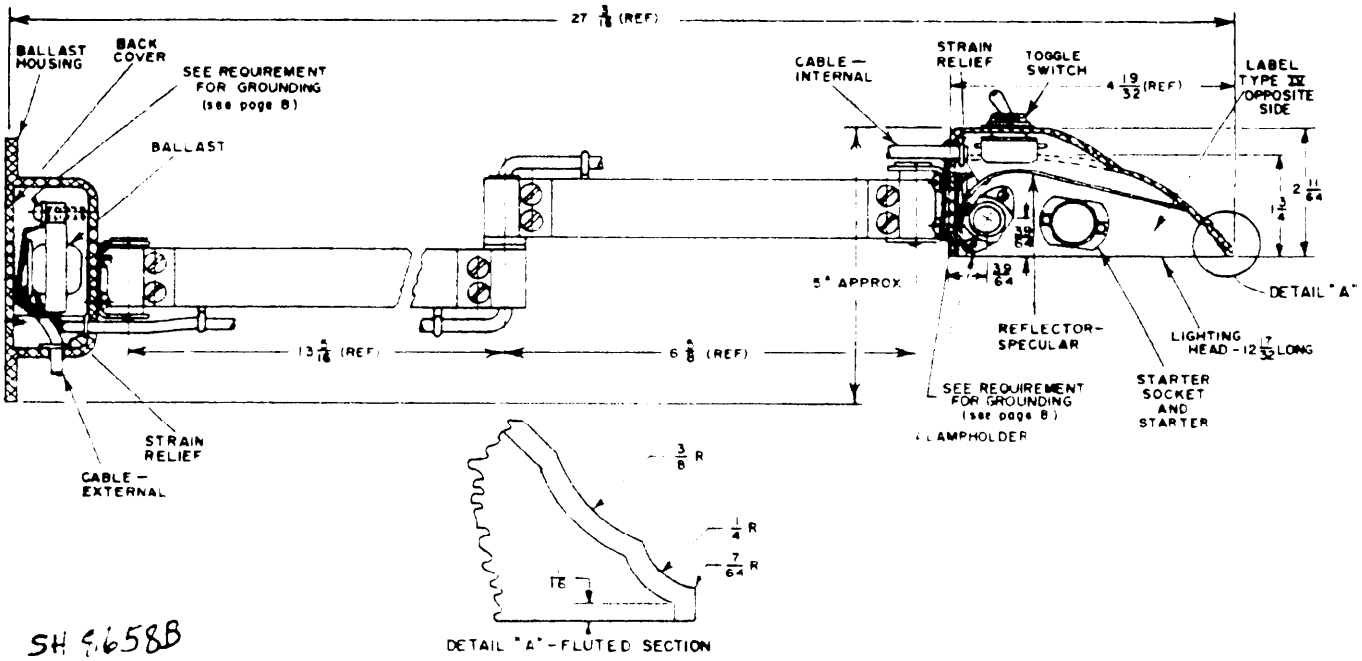


FIGURE 2. Typical section for Symbol 141.2 and 150.2.

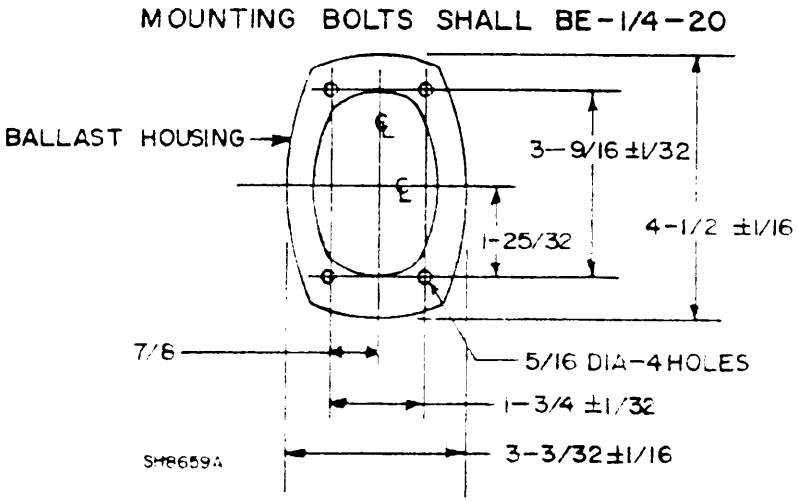


FIGURE 3. Mounting dimensions for Symbol 141.2.

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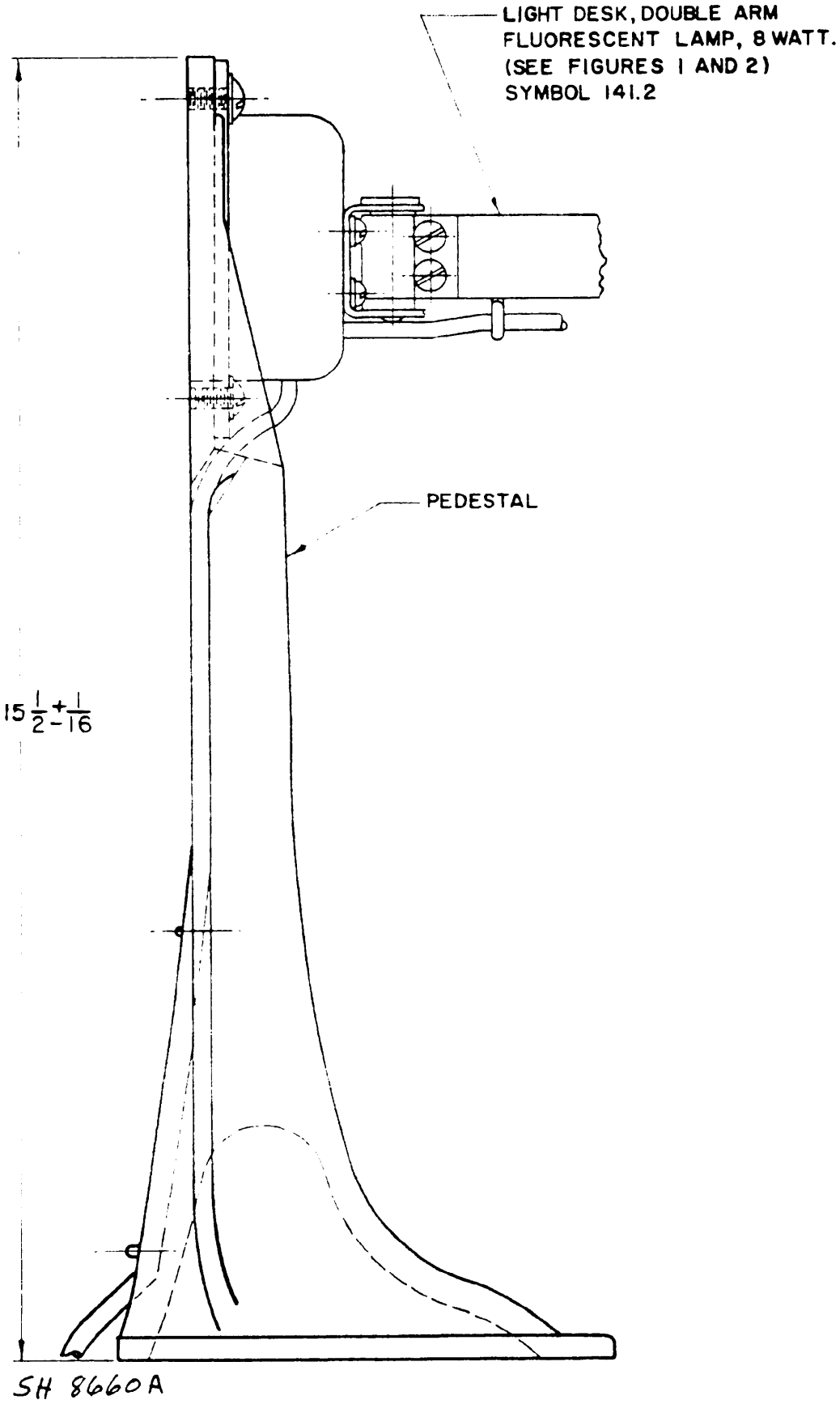
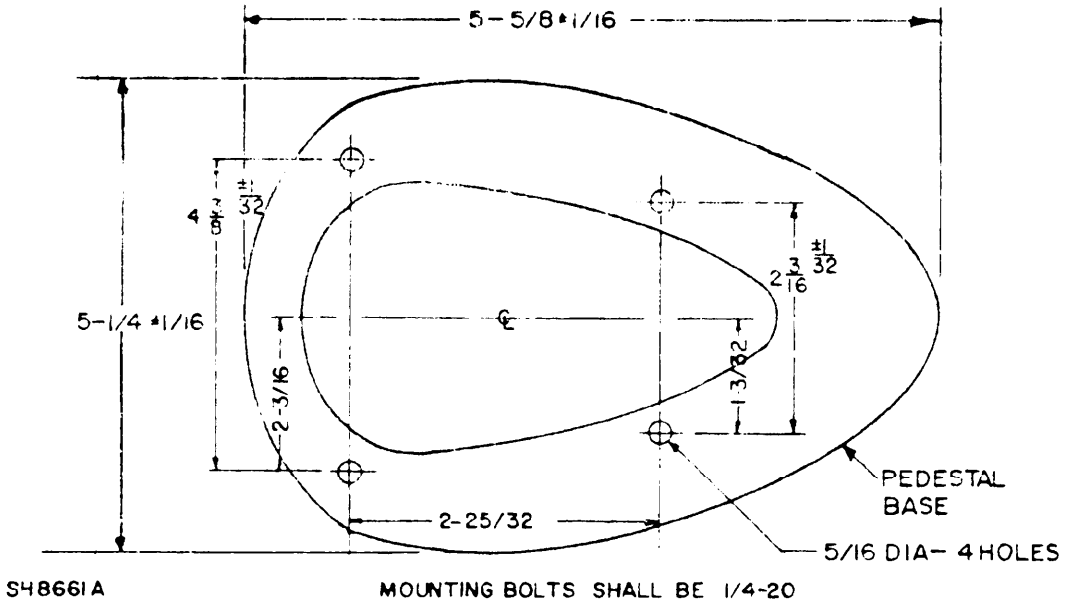
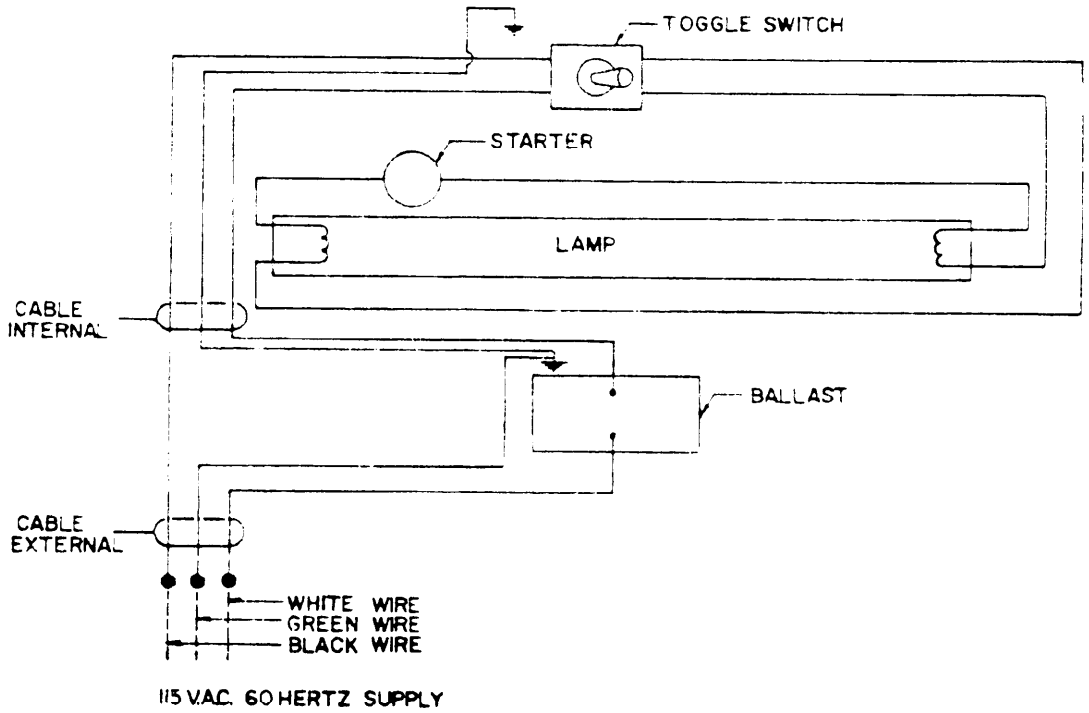


FIGURE 4. Pedestal for Symbol 150.2.

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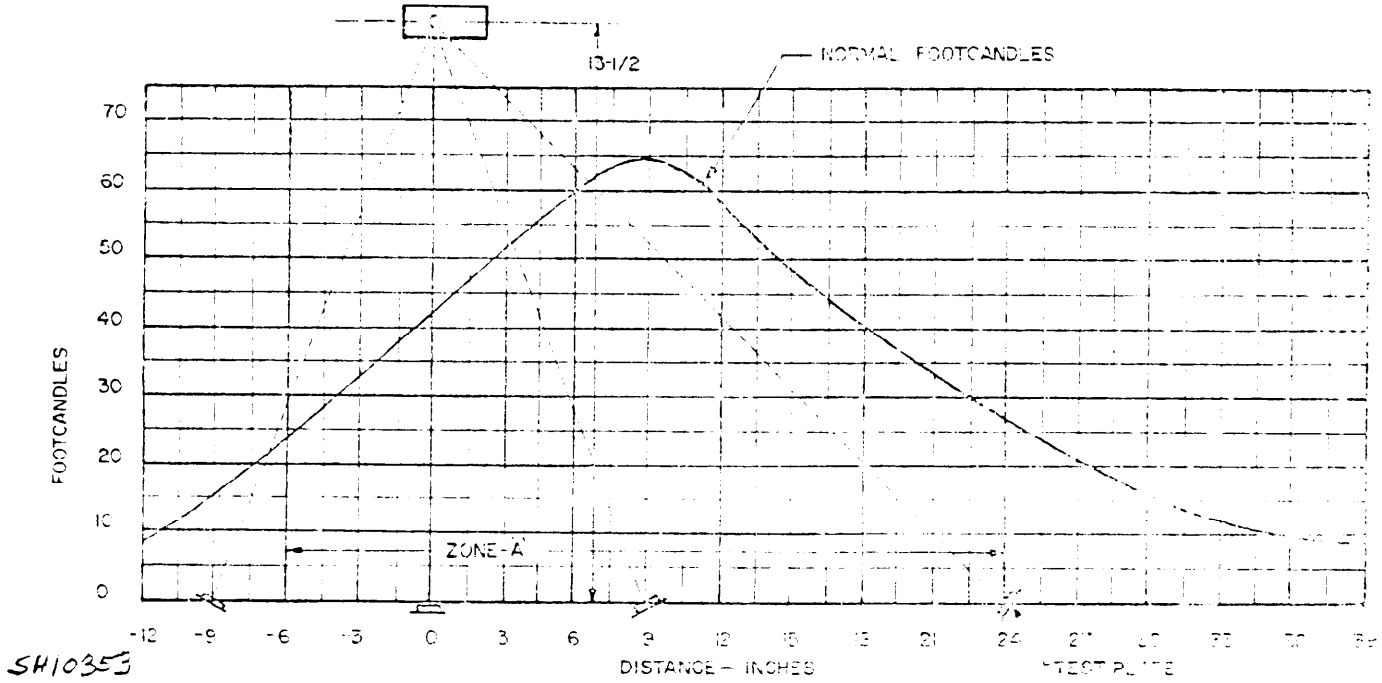


Mounting dimensions for Symbol 150.2.



Wiring diagram for Symbols 111.2 and 150.2.

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SH10353

FOOTCANDLE VALUES WITHIN ZONE "A" SHALL BE NOT LESS THAN 90 PERCENT OF THE VALUES SHOWN HEREON. THESE RESULTS SHALL BE MET AFTER CORRECTION TO 85% LUMENS OUTPUT FOR BOTH LAMP AND BALLAST HAS BEEN MADE.

Photometric test data:

Distance: 13½ inches.
Voltage: 115 volts, a.c., 1 phase
Ambient temperature: 70°F.
Lamp and ballast furnished with fixture shall be used when testing.

FIGURE 7. Footcandle distribution curves for Symbols 141.2 and 150.2.

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REQUIREMENTS:

CLASSIFICATION:
Type: I
Class: 1

SYMBOL NUMBERS:
141.2: Light, desk double arm, bulkhead mounted with one 8-watt fluorescent cool white lamp.
150.2: Light, desk, double arm with pedestal and one 8-watt fluorescent cool white lamp.

DESIGN:
Electrical rating: 8-watts, 120 volts, 60 Hertz.
Dimensions and configuration:
Symbol 141.2: See figures 1, 2 and 3.
Symbol 150.2: See figures 1, 2, 4 and 5.
Distribution curves: Footcandle (see figure 7).
Typical section: See figure 2.
Cable: Internal and external cable shall be type SJT or SVT construction in accordance with J-C-580 with light gray thermoplastic sheath, except size 18 AWG, 3 conductor. For internal and external cable, the green ground leads inside ballast housing shall be terminated with crimp-type terminals and attached under screw and washer. For internal cable, the green ground leads inside the lighting head assembly shall be terminated with crimp-type terminals and connected under round head machine screw and lock-washer. 31 inches of internal, and 10 feet of external cable shall be provided.
Lighting head and supporting arms: The lighting head shall be designed to house the necessary parts, and shall be of a shape that will accommodate the filter assembly shown on MIL-F-16377/48 symbols 121.1 and 121.2. The lighting head shall be made of cast aluminum; minimum thickness shall be 3/32 inch. The supporting arms shall be made of extruded aluminum, minimum thickness shall be 0.074 inch. Swivel pins shall be extruded nylon and hollow corrosion resistant steel tubing. Swivel brackets and wire loops shall be aluminum. Swivel joints shall hold arm firm at all adjustments. Swivel action shall be smooth for all adjustments in the horizontal plane.
Pedestal (Symbol 150.2 only): See figure 4. Material shall be cast aluminum.
Wiring diagram: See figure 6.
Reflector: Specular, prefinished lighting sheet.
Weight:
Symbol 141.2: 3 lbs 9 oz. maximum.
Symbol 150.2: 6 lbs 0 oz. maximum.
Ballast housing: Aluminum, 1/8 inch minimum thickness.
Back cover: Laminated phenolic material.
Strain relief: Strain relief for cables shall be provided as shown on figure 2.
Designation and marking: The following labels shall be furnished in accordance with MIL-F-16377 applied as shown in figures 1 and 3:

(a) One type I.
(b) One type IV. The information to be shown on the type IV label shall be "USE LAMP NSN 6240-00-299-5547 ONLY."

Standard parts: The standard parts shown in table I shall be used in the construction of the fixtures specified herein with no deviation.

QUALITY ASSURANCE: Quality assurance shall be as specified in MIL-F-16377 and as follows:

First article inspection:	See table II.
Quality conformance:	
Comparison inspection:	Same as the first article inspection.
Inspection of product for delivery:	
Examination:	See MIL-F-16377.
Examination for preparation of delivery:	See MIL-F-16377.
Quality conformance tests:	See MIL-F-16377.

GENERAL INFORMATION:

Assignment of Military part numbers and general information is shown on table III. These lights are designed to receive a red and white or blue and white filter shown on MIL-F-16377/48, Symbols 121.1 and 121.2. The filters are not furnished with the lights and must be ordered separately when required. These lights were formerly covered by NAVSHIPS Drawing 9000-S6401-73811 and MS16719.

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3. REQUIREMENTS

TABLE I. Standard parts.

Name	Specification or Military part Number	National Stock Number (NSN)	Remarks
Lamp	W-L-116/3	6240-00-299-5546	Industry Number F8T5/CW
Lamp lock	M16377/40-001	6250-00-399-7039	
Lampholder	MIL-L-970/10	6250-00-377-8590	
Startersocket	MIL-L-970/13	6250-00-295-2738	
Starter	SW, type III of W-S-755	6250-00-299-5962	Commercial Designation: FS5
Ballast	M16377/45-001	6250-00-295-2735	
Switch, toggle DPST	MS16569-1	5930-00-719-9659	
Red and white filter	M16377/48-121.1	6230-00-548-2040	Not furnished
Blue and white filter	M16377/48-121.2		Not furnished

TABLE II. First article and comparison inspection.

Tests	Requirement paragraph	Test paragraph	Remarks
Examination	3.13	4.5	
Operation	3.13	4.8.2	
Dielectric withstanding voltage	3.6.15	4.8.2	
Insulation resistance	3.6.16	4.8.3	
Shock	3.5.12	4.8.7	1/
Vibration	3.5.13	4.8.8	
Dielectric withstanding voltage	3.6.15	4.8.2	
Insulation resistance	3.6.16	4.8.3	
Noise	3.5.14	4.8.9	
Grounding circuit	3.6.13.3	4.8.1.	
Load test			2/
Light output	3.7.2	4.8.6.2	Footcandle
Magnetic permeability	3.4.8	4.8.18	

1/ Lamp breakage or lamp fallout during this test does not constitute failure.
2/ With the desk light installed in the regular mounting position, a vertical load of 15 pounds applied to the swivel joint adjacent to the head assembly shall deflect this joint not more than 3/4 inch.

TABLE III. Part number and general information.

Military Part number M16377/16-	Item name	National Stock Number (NSN)
141.2	Complete light, Symbol 141.2	6230-00-067-4882
150.2	Complete light, Symbol 150.2: Symbol 141.2 and pedestal	6230-00-067-4882 6230-00-336-5876
001	Pedestal	6230-00-336-5876

Revision letters are not used to denote changes due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 6230-N267)

☆U.S. GOVERNMENT PRINTING OFFICE: 1982-505-022/1630

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3.6 Description. The light shall be in accordance with W-L-31[illegible] and as described herein.